Roll No. ....

TEE–501

B. TECH. (EE) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

POWER SYSTEMS–I

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.

(ii) Each question carries 10 marks.

1. (a) Explain Kelvin's Law for the estimation of most economical size of the conductor. (CO1)

OR

(b) A 2-conductor cable 1 km long is required to supply a constant current of 200A throughout the year. The cost of cable including installation is ₹ (20 a + 20) per metre where 'a' is the area of X-section of the conductor in cm^2. The cost of energy is 5P per kWh and interest and depreciation charges amount to 10%. Calculate the most economical conductor size. Assume resistivity of conductor material to be 1.73 $\mu\,\Omega$ cm. (CO1)

P. T. O.

2. (a) Derive the relation between string efficiency and voltage distributions across the disc of suspension type insulators. Assume three discs in the systems. (CO1)

OR

(b) The three bus-bar conductors in an outdoor substation are supported by units of suspension type insulators. Each unit consist of a stack of 3 discs. The voltage across the lowest insulator is 15 kV and that of top unit is 10 kV. Find the bus-bar voltage of the station. (CO1)

3. (a) Discuss the factors, which affect the corona. Explain the following terms : (CO1)

(i) Critical Disruptive Voltage

(ii) Visual Critical Voltage

OR

(b) A d. c. 2-wire system is to be converted into a. c. 3-phase, 3-wire system by the addition of a third conductor of the same cross-section as the two existing conductors. Calculate the percentage additional load which can now be supplied if the voltage between wires and the percentage loss in the line remain unchanged. Assume a balanced load of unity power factor. (CO1)

4. (a) Derive the expression of mechanical sag for the equal support. Also, discuss the effects of ice loading on overhead line conductor. (CO2)

OR

(b) A transmission line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm^2. The tension in the conductor is 2000 kg. If the specific gravity of the conductor material is 9.9 gm/cm^3 and wind pressure is 1.5 kg/m length, calculate the sag. What is the vertical sag ? (CO3)

5. (a) Discuss why high transmission level voltage has been used for transmission of electrical power. (CO2)

OR

(b) A certain 3-phase equilateral transmission line has a total corona loss of 53 kW at 106 kV and a loss of 98 kW at 110 kV. What is the disruptive critical voltage ? What is the corona loss at 113 kV ? (CO3)

50